Advanced Astronomy: Solar System Standards

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AA: Solar System

For 2026–2027, the Geometry and Celestial Sphere objectives have been aligned with the current Geometry and Celestial Sphere I–II packets. Students are also responsible for assigned textbook material and classroom instruction, including Chapter 1; the existing requirements for planetary motion, lunar phases, tidal locking, eclipses, and precession/nutation are retained for separate instructor review of coverage. The Kepler objectives have also been reviewed against the current Kepler I packet; other later sections retain their previous wording pending review. Shared geometry material remains coded as `aa.x.*` because it belongs to the common AA:X strand even when it is used early in AA:SS.

Geometry (shared AA:X prerequisite)

Topic	I am able to…	Code (`aa.x.euclidean_geometry.`)
Triangles	… use sines, cosines, tangents, and Pythagoras' Theorem to solve right triangles quantitatively.	`triangles.010`
	… identify which geometric tool is appropriate in a given applied problem.	`triangles.020`
	… apply right-triangle geometry to real astronomical estimation problems involving lengths, distances, and angles.	`triangles.030`
	… use geometric arguments to explain special values of sine, cosine, and tangent, the relation between tangent and sine/cosine, and the relations between complementary angles.	`triangles.040`
Radians	… explain what a radian measures geometrically.	`radians.010`
	… use radians to determine arc lengths and related quantities.	`radians.020`
Units	… convert among decimal degrees, degrees–arcminutes–arcseconds, and radians.	`units.010`
	… define parsecs (pc) and astronomical units (au), and convert among those units and meters.	`units.020`
Parallax	… use a diagram of annual stellar parallax to distinguish the total apparent shift from the parallactic angle and the full observing baseline from the one-au baseline in the right triangle, and use this geometry to determine distance.	`parallax.010`
	… relate astronomical distances to light-travel times and express those times in appropriate units.	`parallax.020`
Small-angle approximations	… derive an exact angular size from right-triangle geometry and compare it with the arc-length approximation for the same object.	`small_angles.010`
	… explain geometrically why a short arc approaches its chord and why sine, the angle in radians, and tangent approach one another for small angles.	`small_angles.020`
	… calculate percentage discrepancies between exact and approximate results and determine whether a small-angle approximation meets a stated accuracy requirement.	`small_angles.030`
Measurement uncertainty	… use the bounds on a measured angle to estimate the resulting interval in an inferred length, and distinguish that uncertainty interval from the percentage discrepancy relative to a reference value.	`uncertainties.010`

Celestial Sphere, Prominent Stars, Constellations

Topic	I am able to…	Code (`aa.ss.celestial_sphere.`)
Diurnal motion / rotation / revolution	… explain what diurnal motion is.	`diurnal_motion.010`
	… clearly distinguish rotation from revolution.	`diurnal_motion.020`
	… explain what prograde and retrograde motion are.	`diurnal_motion.030`
	… explain that the orbital motion of the major Solar System bodies is generally prograde.	`diurnal_motion.040`
	… explain that the rotational motion of most major Solar System bodies is generally prograde.	`diurnal_motion.050`
	… use my understanding of these motions to predict the phase and sky position of the Moon from given observational information.	`diurnal_motion.060`
	… explain what it means for the Moon to be tidally locked to the Earth, and relate this to its rotation and revolution.	`diurnal_motion.070`
Ecliptic / equator / seasons	… explain what the ecliptic and ecliptic plane are.	`ecliptic_seasons.010`
	… explain what the Earth's equatorial plane is.	`ecliptic_seasons.020`
	… relate seasonal observations to the inclination of the Earth's rotation axis.	`ecliptic_seasons.030`
	… explain why summers are warm and winters are cold.	`ecliptic_seasons.040`
	… explain what the equinoxes and solstices are, when they occur, and where the Earth is in its orbit when they occur.	`ecliptic_seasons.050`
	… relate the Sun's changing declination and daily path to its rising and setting directions, noon altitude, and length of daylight at a specified observing latitude and season.	`ecliptic_seasons.060`
	… describe an analemma as the pattern of the Sun's positions observed at the same clock time over a year.	`ecliptic_seasons.070`
Eclipses	… give geometric explanations of solar and lunar eclipses.	`eclipses.010`
	… explain what is happening physically when solar and lunar eclipses occur.	`eclipses.020`
	… explain when eclipses can occur and when they cannot.	`eclipses.030`
	… explain the phases of the Moon during eclipses.	`eclipses.040`
	… determine the maximum lengths of eclipses.	`eclipses.050`
Celestial coordinates	… explain position on Earth in terms of latitude and longitude.	`coordinates.010`
	… explain the horizontal coordinate system using altitude and azimuth, identify the horizon and zenith, and state the reference direction and sense in which azimuth is measured.	`coordinates.020`

Topic	I am able to…	Code (`aa.ss.celestial_sphere.`)
	… locate specific objects in the sky using an appropriate coordinate system.	`coordinates.030`
	… define right ascension and declination using the celestial equator, celestial poles, and vernal-equinox direction, and interpret their units and coordinate grids.	`coordinates.040`
	… explain how horizontal coordinates depend on observing location and time, and how the equatorial grid moves relative to the horizon as Earth rotates.	`coordinates.050`
	… derive and apply the relation between latitude, solar declination, and the Sun's altitude at local solar noon, including equinox and solstice cases.	`coordinates.060`
	… convert geographic coordinates between decimal and degrees–arcminutes–arcseconds form, and calculate surface distances from central angles along great-circle arcs on a spherical Earth.	`coordinates.070`
Observation and instruments	… construct and use a simple altitude-measuring instrument, estimate and justify its angular uncertainty, and explain how its design could be improved.	`observations.010`
	… record an object's altitude and azimuth with the observing location and date/time, and use a chart or sky map for the same observing conditions to support an identification.	`observations.020`
	… compare coordinate estimates or measurements with a reference and assess whether their differences are consistent with the claimed uncertainty.	`observations.030`
Star charts and constellations	… distinguish a constellation from an asterism and use recognizable star patterns and coordinate grids to locate objects, including Polaris and the neighborhood of the equatorial coordinate origin.	`star_charts.010`
	… trace the Sun's annual path through the zodiacal constellations on a star chart.	`star_charts.020`
	… construct a planisphere appropriate to an observing latitude and use its date/time settings and horizon window to locate visible star patterns.	`star_charts.030`
Astranatomy	… calibrate the angular sizes of my own finger, fist, and outstretched hand from measured widths and eye-to-hand distances, and explain their use for rough observation.	`astranatomy.010`
	… estimate angular sizes of objects and angular distances on the celestial sphere.	`astranatomy.020`

Topic	I am able to…	Code (`aa.ss.celestial_sphere.`)
Precession / nutation	… explain what precession and nutation are.	`precession_nutation.010`
	… explain the influence of the Moon on the Earth's precession.	`precession_nutation.020`

Aristarchus of Samos

Topic	I am able to…	Code (`aa.ss.aristarchus.`)
History	… explain who Aristarchus was and what he believed about the Sun, planets, and stars.	`history.010`
	… explain some of the reasons Aristarchus' model was rejected by his contemporaries, including arguments involving parallax, wind, and the motion of objects on Earth.	`history.020`
Sun–Earth–Moon	… apply Aristarchus' logic to modern observations to determine relative sizes of the Sun, Earth, and Moon.	`sun_earth_moon.010`
	… apply Aristarchus' logic to modern observations to determine relative sizes of the Sun–Earth and Earth–Moon distances.	`sun_earth_moon.020`
	… apply Eratosthenes' method to modern observations to determine the size of the Earth.	`sun_earth_moon.030`

Kepler's Laws

Assessment scope (2026–2027): Related objectives remain grouped by topic. Unit 2 covers Kepler I geometry and representations and the Third Law applications developed with Dimensional Analysis and Astrodynamics I. Unit 2 also includes the qualitative equal-areas rule and orbital-speed variation in the assigned textbook sections. The follow-up Unit 3 develops the angular-momentum and energy explanations of orbital motion. Unit-specific assessment lists will identify the VLOs, or components of broader VLOs, being tested. Distinctions among anomalies and anomaly-conversion or Kepler-equation derivations are enrichment and will not be tested.

Topic	I am able to…	Code (`aa.ss.kepler.`)
Copernicus	… explain that Copernicus openly argued that the Earth and other known planets orbit the Sun.	`copernicus.010`
	… explain why the Copernican model did not automatically outperform the Ptolemaic model while both still used circular orbits.	`copernicus.020`
	… state that Copernicus calculated the orbital periods of the planets.	`copernicus.030`

Topic	I am able to…	Code (`aa.ss.kepler.`)
Tycho / Kepler	… explain who Tycho Brahe and Johannes Kepler were.	`tycho_kepler.010`
	… give a rough explanation of how Kepler used Tycho's data and Copernican ideas to obtain his first two laws.	`tycho_kepler.020`
	… explain Kepler's First Law and how it corrected the chief geometric flaw in the Copernican model.	`tycho_kepler.030`
	… explain Kepler's Second Law.	`tycho_kepler.040`
Ellipses	… construct ellipses geometrically.	`ellipses.010`
	… represent the same ellipse in Cartesian, parametric, and focus-centered polar forms, keeping track of the origin and angle convention.	`ellipses.020`
	… relate semi-major axis, semi-minor axis, focal distance, eccentricity, periapsis, and apoapsis, and calculate these quantities from one another.	`ellipses.030`
Conic sections	… use a cone-and-plane model to distinguish circles, ellipses, parabolas, and hyperbolas and explain how the cutting plane determines the section.	`conics.010`
Applications of Kepler's Laws	… state Kepler's First Law.	`applications.010`
	… explain Kepler's Second Law in terms of motion on an ellipse and relate it to a conservation law, namely angular momentum.	`applications.020`
	… explain what Kepler's Second Law means for orbital speed near perihelion and aphelion.	`applications.030`
	… apply Kepler's Third Law to predict orbital periods or orbital radii from a table of known values.	`applications.040`
	… generalize Kepler's Third Law to other systems, such as the Galilean moons, and explain what changes and why.	`applications.050`

Solar System I

Topic	I am able to…	Code (`aa.ss.solar_system1.`)
Features / structure	… describe the gross features of the Solar System in terms of the Sun, the four terrestrial planets, the four giant planets, the main asteroid belt, the Kuiper belt, and the Oort cloud.	`features_structure.010`
	… explain the relative sizes of the major Solar System bodies.	`features_structure.020`
	… explain the relative distances between the major Solar System bodies.	`features_structure.030`

Solar System II

Topic	I am able to…	Code (`aa.ss.solar_system2.`)
Formation theory	… explain nebular theory in terms of interstellar matter (ISM), giant molecular clouds (GMCs), dust, radiative cooling, Jeans instability, cloud collapse and fragmentation, angular momentum, spin-up, flattening, frictional heating, the T Tauri phase, and gas expulsion.	`formation_theory.010`
	… explain qualitatively what angular momentum is and how its conservation leads to spin-up and flattening of the protoplanetary disk.	`formation_theory.020`
	… explain what accretion is and how planetesimals form.	`formation_theory.030`
	… explain how heating of the collapsing protostellar bulge contributed to planetary differentiation.	`formation_theory.040`
	… explain what the asteroid belt is and why it exists.	`formation_theory.050`
	… explain the roles of the gas giants in the formation of the Kuiper belt and Oort cloud.	`formation_theory.060`
	… describe with proper vocabulary what is happening in numerical simulations of GMC collapse and fragmentation.	`formation_theory.070`
	… describe the temperature profile of the proto-solar system, identify the ice line, describe the formation of refractory materials, and use this to explain planetary differentiation.	`formation_theory.080`
	… describe gravitational assists by appealing to conservation of linear momentum, and relate them both to space travel and to Solar System structure.	`formation_theory.090`

Solar System III

Topic	I am able to…	Code (`aa.ss.solar_system3.`)
Quantitatively	… apply the Virial Theorem to determine conditions for the stability of a warm, self-gravitating molecular cloud and the onset of Jeans instability.	`quantitative.010`
	… calculate the Jeans mass M_J or critical temperature T_c from thermodynamic stability arguments for a cloud fragment.	`quantitative.020`
	… calculate the free-fall time $t_{\rm ff}$ for the collapse of a giant molecular cloud from its initial density.	`quantitative.030`
	… calculate how the temperature of the central region increases as a function of the size of a collapsing cloud.	`quantitative.040`
	… construct a realistic model of the temperature profile of the proto-solar system and identify the ice line.	`quantitative.050`
	… calculate the change in velocity, Δv, of a small body in a one-dimensional gravitational assist.	`quantitative.060`

Sol

Topic	I am able to…	Code (`aa.ss.sol.`)
Structure	… describe the Sun in terms of its total mass, radius, density, luminosity, and general observational character.	`structure.010`
	… describe the structure of the Sun in terms of the core, radiation zone, convection zone, photosphere, chromosphere, and corona.	`structure.020`
	… describe the Sun's temperature structure and differential rotation.	`structure.030`
	… explain hydrostatic equilibrium and zonal flow in the context of the Sun.	`structure.040`
Solar wind / weather	… describe the basics of radiation using the Planck curve, Wien's Law, and the Stefan–Boltzmann law.	`solar_wind_weather.010`
	… calculate the Sun's surface gravity, escape speed, luminosity, and surface temperature.	`solar_wind_weather.020`
	… use those physical quantities to reason about solar mass loss.	`solar_wind_weather.030`
	… describe the Sun's magnetic field and its implications for polarity, sunspots, and the solar cycle.	`solar_wind_weather.040`

Topic	I am able to…	Code (aa.ss.sol.)
	… describe solar flares, coronal mass ejections (CMEs), and space weather, including the hazards they pose for Earth.	solar_wind_weather.050
Fusion	… describe how energy is generated in the core using basic nuclear particle physics, the mass–energy relation, and the proton–proton chain.	fusion.010
	… identify the roles of protons, neutrons, electrons, neutrinos, and nuclei in solar fusion processes.	fusion.020
	… calculate the rate of solar energy generation in an appropriate simplified model.	fusion.030

Jupiter I

Topic	I am able to…	Code (aa.ss.jupiter1.)
Bulk properties / rotation	… describe Jupiter in terms of its total mass, size relative to Earth and the Sun, average density, and the implications of that density for composition.	bulk_properties_rotation.010
	… describe Jupiter's rapid rotation rate and its major observational consequences.	bulk_properties_rotation.020
	… estimate and compare Jupiter's surface gravity and equatorial rotation speed using order-of-magnitude and scaling arguments.	bulk_properties_rotation.030
Rotating frames	… describe the Coriolis pseudo-force and how it gives rise to zonal flow.	rotating_frames.010
	… describe the centrifugal pseudo-force and how it relates to flattening and oblateness.	rotating_frames.020
	… use the right-hand rule to determine the directions of relevant pseudo-forces in a rotating frame.	rotating_frames.030
Atmosphere / weather	… describe the gross features of Jupiter's atmosphere in terms of bright zones, dark belts, large-scale storms, east–west zonal flows, and differential rotation.	atmosphere_weather.010
	… explain how rapid rotation and pseudo-forces organize atmospheric motion and lead to alternating bands with opposite flow directions.	atmosphere_weather.020
	… identify the role of the Coriolis pseudo-force in shaping Jupiter's banded structure and large-scale atmospheric circulation.	atmosphere_weather.030
Interior / composition	… describe what is and is not understood about the interior structure of Jupiter.	interior_structure.010

Topic	I am able to…	Code (`aa.ss.jupiter1.`)
	… describe Jupiter's internal structure qualitatively, including outer gaseous layers, deeper fluid regions, and a dilute or uncertain core.	`interior_structure.020`
	… describe what a phase diagram represents and use the phases of hydrogen to draw conclusions about Jupiter's interior.	`interior_structure.030`
	… explain why hydrogen is expected to behave like a metal at depth in Jupiter because of the extremely high pressures there.	`interior_structure.040`
	… explain why Jupiter radiates more energy than it receives from the Sun, and relate this to Kelvin–Helmholtz contraction and the slow release of gravitational energy.	`interior_structure.050`
Magnetosphere	… describe the origin of Jupiter's strong magnetic field in terms of rapid rotation and electrically conducting material in the deep interior.	`magnetosphere.010`
	… describe the scale and significance of Jupiter's magnetosphere, including trapped charged particles, radio emission, and the radiation environment near the moons.	`magnetosphere.020`

Jupiter II

Topic	I am able to…	Code (`aa.ss.jupiter2.`)
Moons and rings	… identify the Galilean moons Io, Europa, Ganymede, and Callisto.	`moons_rings.010`
	… explain what orbital resonance is and how it applies to the Galilean moons.	`moons_rings.020`
	… describe the Amalthean group—Metis, Adrastea, Amalthea, and Thebe—and locate them relative to Jupiter, its rings, and the Galilean moons.	`moons_rings.030`
	… describe Jupiter's moon-and-ring system in terms of its major subdivisions and the relationships among rings, inner moons, and Galilean moons.	`moons_rings.040`
	… describe the Galilean moons as a system, including their relative sizes, densities, orbital distances, and the trend of decreasing density with increasing distance from Jupiter.	`moons_rings.050`
	… explain the analogy between the Galilean system and a "mini solar system."	`moons_rings.060`
	… describe how the moons interact with Jupiter through gravity, tides, resonance, and the magnetic-field environment.	`moons_rings.070`

Topic	I am able to…	Code (`aa.ss.jupiter2.`)
Rotating frames / pseudo-forces	… describe motion in a rotating reference frame in terms of centrifugal and Coriolis pseudo-forces.	`rotating_frames_pseudo_forces.010`
	… determine when pseudo-forces must be included and when they should not.	`rotating_frames_pseudo_forces.020`
	… describe and identify the five Lagrange points of a two-body system and explain how they relate to Jupiter's Trojan asteroids.	`rotating_frames_pseudo_forces.030`
	… analyze the stability of Lagrange points, including the distinction between unstable, absolutely stable, and dynamically stable equilibrium points.	`rotating_frames_pseudo_forces.040`
	… predict qualitatively the motion of objects displaced slightly from the neighborhood of a Lagrange point.	`rotating_frames_pseudo_forces.050`